ISTA Safaris Android Build Note (Flutter)

Nov 7, 2026 · @Rooney Odhiambo

1. Purpose and how to use this note

Build the ISTA Safaris Android app in Flutter, with the Django API and admin it depends on, so that it looks like the web design in `ista-safaris.zip` and follows the stack in the project budget (25 October 2026).

Give Claude Code three inputs, placed in one repository:

Input	Where to put it	What it decides
This note, exported as Markdown	`CLAUDE.md` at the repository root (Claude Code reads it automatically)	How to work, what to build, in what order
`ISTA_Safaris_Full_Project_Budget.docx`	`docs/budget/`	Scope, stack, phases and what is out of scope
`ista-safaris.zip`, unzipped	`reference/web/`	Look, layout, copy, colours, fonts, images

Order of authority: this note first, then the budget, then the web reference. Where two of them disagree, stop and ask the owner.

The web reference is a design source, not code to port. It is built with React, Supabase and public demo map services; none of these go into the Android app or the production backend.

Scope is Android first, as the budget sets out. iOS, the AI assistant, bookings and payments are Phase 2 items with their own funding decision and are out of scope here.

First message to send Claude Code:

```
Read CLAUDE.md, docs/budget/ and reference/web/ in full. Do not write any
code yet.
Reply with: (1) the proposed repository file tree, (2) every package you plan
to use with its licence and purpose, (3) any conflicts or open questions you
found. Wait for my approval.
```

2. Working rules for Claude Code

Work in small, approved steps: no code is written for a phase until the owner has approved its file tree, and no phase starts until the previous one is signed off.

1. **File tree first.** At the start of every phase, list the files you will create or change and any packages you will add. Wait for approval.
2. **Sign-off between phases.** At the end of each phase, stop and report what was built, how you checked it (tests, screenshots) and what is left. Wait for an explicit "approved" before moving on.
3. **Packages need a reason.** For every package give its name, version, licence, purpose and what data it sends off the device. Ask before adding anything under GPL or AGPL, or anything that collects analytics.
4. **Never invent content.** No made-up ratings, reviews, counts, distances, drive times, file sizes, weather or claims. Show real data from the API, or hide the element. Section 7 lists the cases in the design.
5. **No secrets in the repository.** Keys go in `.env` (backend) and `--dart-define-from-file` (app). Commit only `.env.example` files.
6. **Ask, do not guess.** When this note, the budget and the design disagree, or a decision in section 9 is still open, stop and ask.
7. **Checks before "done".** Run `flutter analyze`, `flutter test` and the Django test suite, and fix every failure before reporting a phase complete.
8. **Keep a log.** Record each phase's decisions, deviations from the design and open questions in `docs/PROGRESS.md`.
9. **Git hygiene.** One branch per phase, small commits with clear messages, no force pushes, no rewriting history.

3. Tech stack

The budget fixes the core stack: a Flutter Android app, a Django REST API with the Django admin as the content portal, PostgreSQL, and an on-device SQLite cache for offline packs (budget sections 1, 2 and 6). The rows marked *recommended* are Claude Code's starting choice and need the owner's approval in Phase 0.

Layer	Use	Basis
Mobile app	Flutter (latest stable) and Dart, Android only	Budget
Android target	`targetSdk 36`, `minSdk 24`	Play requires API 36 for new apps from 31 Aug 2026 (source); `minSdk` is recommended
State and navigation	Riverpod, go_router	Recommended
HTTP client	dio, with token refresh and retry	Recommended
Local cache	SQLite through drift: places, safety, routes, saved items, pack index	Budget (SQLite local cache)
Token storage	flutter_secure_storage	Recommended
Maps	flutter_map with tiles from a licensed provider: MapTiler Cloud or Google Maps Platform	Budget; offline tile rights must be confirmed first
Location	geolocator, foreground only	Budget (GPS)
Audio guides	just_audio	Budget (audio player)
Languages	flutter_localizations and ARB files, English and Kiswahili first	Budget (localization ready)
Icons	Lucide icon set (ISC licence) and three custom icons from the design	Design
Fonts	DM Serif Display and Manrope, bundled TTF files (SIL Open Font License)	Design
Error reporting	Sentry or similar, off until the user agrees	Budget (monitoring)

Layer	Use	Basis
Backend	Django, Django REST Framework, Django admin	Budget
Database	PostgreSQL, managed, with backups	Budget
Auth	Email and password, JWT access and refresh tokens (djangorestframework-simplejwt)	Recommended
Media	S3-compatible storage and CDN for photos, audio and offline packs (django-storages)	Budget
Weather	Open-Meteo, called by the backend and cached	The free API is non-commercial only; commercial use needs a subscription (source)

Do not carry over from the web reference: React, Vite, Leaflet, Supabase, the public OSRM demo router or the public Nominatim geocoder. They are fine for a design preview and wrong for a production app.

4. Design reference

Match the web design's look on a phone, starting from its mobile layout (the CSS rules under `max-width: 960px` and `max-width: 560px`), not its desktop layout. The web top navigation becomes an Android bottom navigation bar in the same colours.

Where to look in `reference/web/`:

Path	What it holds
`src/styles/global.css`	Every colour, font size, spacing, radius, shadow and the mobile breakpoints
`src/pages/`	Page structure and copy for Explore, Directions, Sign in, Sign up and the legal pages
`src/components/`	Each component's layout, copy and states (home, directions, auth, layout)

Path	What it holds
`src/components/icons.tsx`	Three custom icons to redraw: route, sparkle, thin arrow
`src/data/places.ts`, `src/data/content.ts`	Six seed places and the design copy marked `CHECK`
`public/images/`, `public/icons/logo.png`	Photos and logo captured from a screen recording, screen resolution only: placeholders

Colour tokens

Token	Hex	Used for
background	#FCFBF4	Page background
card	#FDFCF6	Cards, sheets, map overlays
field	#FFFFFC	Input fill
ink	#183828	Headings and body text
forest	#23523B	Primary buttons, download card, directions band, active chip
forestDark	#17392B	Ask ISTA icon circle
terracotta	#BF553A	Map pins, secondary buttons, large eyebrows and links
gold	#E8C47C	Italic accent words and icons on forest
goldButton	#EACD85	Download region button
goldCard	#DFB960	Ask ISTA card
sand	#F2EFE2	Smart suggestion card, icon buttons, selected list row
line	#E5E1D8	Borders
lineSoft	#EBE7DF	Dividers
muted	#7D7E73	Secondary text in the design

Token	Hex	Used for
online	#5AB36C	Status dot

Three design colours fail WCAG AA (4.5:1) for small text, so use darker values for text under 18 sp and keep the design values for icons, pins and large text:

Pair	Design ratio	Use instead for small text	New ratio
Terracotta on background	4.42	#B04B32	5.19
Terracotta on sand	3.98	#B04B32	4.67
Muted on background	3.97	#6B6C61	5.13
Navigation grey #6B786D on background	4.47	#6B6C61	5.13

Type, shape and icons

- **DM Serif Display**, regular and italic: every heading, place name and card title. The second line of hero titles is italic in gold: "Go further.", "heading next?", "Know more.", "here."
- **Manrope** 400 to 800: body, labels and buttons. Eyebrows are Manrope 800, uppercase, letter spacing 0.2 em. Buttons are Manrope 800, uppercase, letter spacing 0.06 to 0.1 em.
- Bundle both fonts as TTF files in `assets/fonts/` so text renders offline. Do not fetch fonts at runtime.
- Start from the mobile sizes in the CSS (hero title 42/44, section titles 40/44, sign in and sign up titles 40/44) and map them to a Flutter `TextTheme`. Support system text scaling up to 200% without clipping.
- Radius: 6 for buttons and inputs, 12 for cards, 16 for large containers, fully rounded for chips and pills. Touch targets at least 48 dp.
- Icons: the Lucide set for standard icons; redraw the route, sparkle and thin arrow icons from `icons.tsx` as SVG assets.
- Photos in production come from the backend with alt text and credit. Only the logo is bundled in the app; replace it with the original high resolution file.

Build a debug-only **design gallery** screen showing every token and component, and compare it with screenshots of the web reference at 390 px wide before building screens.

5. Screens and features

The app has four bottom tabs (Explore, Map, Directions, Profile) plus the screens below. Screens marked *new* are not in the web design and must be built from the same design system, then shown to the owner as a mock-up before coding.

Screen	Source	What it does
Explore (home)	Design: Explore page	Header with logo and status pill; hero with weather line, title, search and voice search; Safety center button; places list with map; greeting; smart suggestion; download card; Explore nearby with filter chips and cards; Discover list; Ask ISTA card
Map	Design: map card on Explore	Full-screen map of the region with numbered pins and the selected place card ("View place")
Place detail	*New*	Photos, kind, name, description, audio guide, safety notes for that place, credits, Save, Navigate
Directions	Design: Directions page	Current location (with a "use my location" button), destination with place suggestions, route on the map, distance, time and step list
Safety center	Design: hero button (dialog)	Emergency numbers 999 and 112, park rules, approved safety notices; works offline
Ask ISTA	Design: Ask ISTA card	Answers drawn only from verified content in the app (places, safety, FAQs). The AI assistant is Phase 2
Offline packs	*New*	Region packs with real size, version and date; download, update, delete, progress and storage used
Sign in	Design: Sign in page	Email and password; forgot password link (*new* reset screen)

Screen	Source	What it does
Sign up	Design: Sign up page	Full name, optional phone, email, country, preferred language, password with show and hide, confirm password, one unticked consent box for age (18 or over, or guardian permission) and the terms and privacy policy
Profile	*New*	Edit details, language, saved places, offline packs, privacy choices, legal pages, sign out, delete account
Legal	Design: legal pages	Privacy policy, terms, refund policy, data and storage notice, business details, delete my data
First launch	*New*	Short privacy notice with equal "Allow" and "No thanks" buttons for optional error reporting

Rules for design elements whose data does not exist yet:

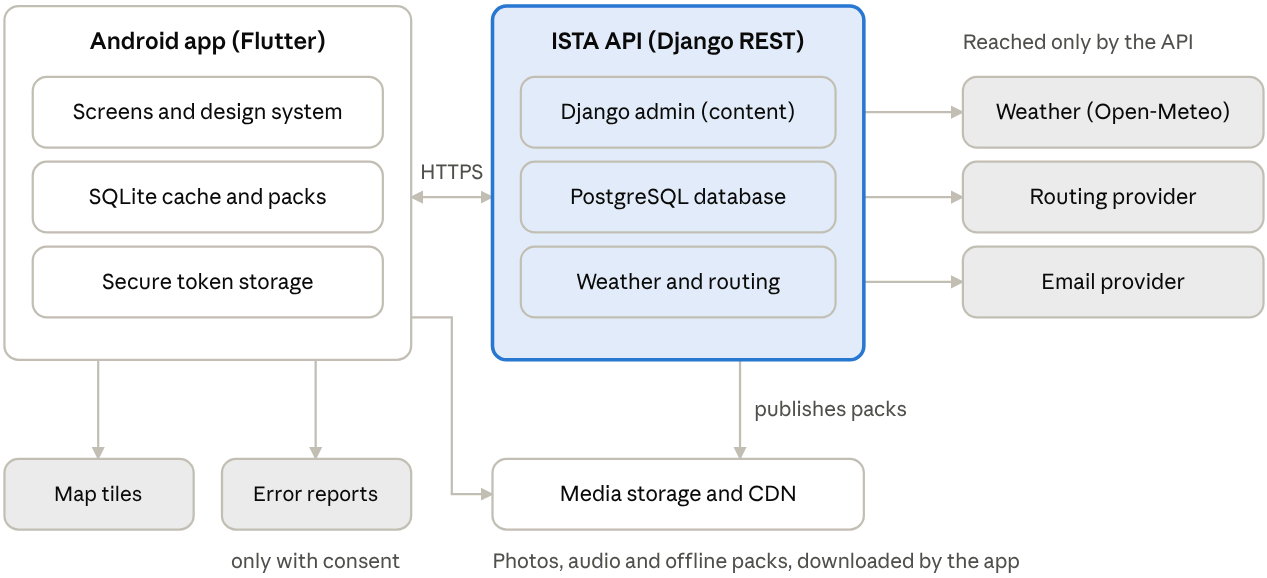
- **Star ratings**: keep the space in the card layout but show a rating only when the API returns one built from real reviews. The MVP has no reviews, so ratings stay hidden.
- **Discover counts** ("64 sightings", "24 lodges"): compute from published records per category. Hide a category with no records.
- **Distances and drive times**: compute from the user's location and the routing service. Without a route, show straight-line distance labelled as such and no drive time.
- **Smart suggestion**: pick the nearest published place to the user and use that place's own tip text. Without location permission, show no suggestion.
- **Download card**: size and "updated" date come from the pack manifest, never fixed text like "482 MB".
- **Status pill**: "Offline ready" only when the current region pack is on the device; "Offline mode" with no connection; otherwise no pill.
- **Weather line**: live data, hidden when offline or when the weather service fails.
- **Ask ISTA card**: use "Your local guide." until a multilingual assistant exists; "in any language" is a claim the MVP cannot support.
- **Nearby cards**: "Mudanda Rock" and "Lion Ridge" come from the design file; publish them only if the owner confirms they are real, verified places.

Seed data for the prototype: the six places in `src/data/places.ts` (Voi, Aruba Dam, Taita Hills, Mzima Springs, Lake Jipe, Lumo Sanctuary). Their coordinates are approximate and must be verified before release.

6. Data, backend and integrations

The app talks to four things: the ISTA API, the media CDN that serves photos, audio and packs, the map tile provider and, only with the user's consent, the error reporting service. Weather, routing and email are reached only through the API, so their keys stay on the server and providers can be swapped.

The app reaches weather, routing and email only through the ISTA API



System architecture · app, API and outside services

Shaded boxes are outside providers; the highlighted API is the only part that holds third-party keys other than the restricted map key.

Repository layout (to confirm in Phase 0)

```
ista-safaris/
  CLAUDE.md            this note
  mobile/              Flutter app (Android)
  backend/             Django project: api, content, accounts, packs apps
  docs/                budget, PROGRESS.md, THIRD_PARTY.md, API notes
  reference/web/       unzipped design reference (read only)
```

Data model

Model	Key fields
Region	name, slug, map bounds, default zoom
Place	region, slug, name, kind, category (wildlife, landmark, stay, culture), tagline, description, tip, latitude, longitude, published, verified by, verified at
PlaceImage	place, file, alt text (required), credit, licence, order
AudioGuide	place, language, file, duration, transcript, narrator credit
SafetyNotice	region or place, title, body, severity, approved by, approved at, valid until
CuratedRoute	place, geometry, distance, duration, notes, approved by
OfflinePack	region, version, size in bytes, manifest file, checksum, built at, includes map tiles (yes or no)
Profile	user, full name, phone (optional), country, language, age confirmed at, terms version and accepted at, privacy version
SavedPlace	user, place, created at
DeletionRequest	user, requested at, status, completed at

Admin rules: a place or safety notice cannot be published without a verifier and date; an image cannot be published without alt text, credit and licence. This puts the budget's verification and content rights lines into the workflow.

API (versioned under `/api/v1/`, OpenAPI schema published)

- Auth: `POST auth/register`, `POST auth/token`, `POST auth/token/refresh`, `POST auth/password-reset`, `POST auth/verify-email`
- Account: `GET` and `PATCH me`, `DELETE me` (deletes the account and personal data, confirms by email), `GET`, `POST` and `DELETE me/saved`

- Content: `GET regions`, `GET regions/{slug}/places?category=`, `GET places/{slug}`, `GET places/{slug}/routes`, `GET safety?region=`, `GET legal/{document}`
- Offline: `GET packs`, `GET packs/{id}/manifest`
- Services: `GET weather?region=` (cached for 30 minutes), `GET route?from=&to=` (proxy to the chosen routing provider)

Offline packs

- A pack is a JSON manifest listing every record, photo, audio file and (only if licensed) map tile archive, each with URL, size and SHA-256 checksum.
- The app downloads into app storage, resumes interrupted downloads, verifies every checksum, writes records to SQLite and swaps in a new version only when complete.
- The size shown to the user is the manifest total. Packs are downloaded only when the user asks; offer a "Wi-Fi only" option.
- Until offline map rights are confirmed, packs carry content, photos and audio but not map tiles, and the app says so.

Outside services

Service	Choice	Notes
Map tiles	MapTiler Cloud (budget lists Flex at US$30 a month) or Google Maps Platform	Restrict the key to the app's package name and signing certificate
Routing	Open decision (section 9)	Curated, admin-approved routes take priority inside parks
Weather	Open-Meteo via the API	Commercial use needs a subscription
Email	Transactional only: verification and password reset	Provider to choose
Error reporting	Sentry or similar	Off until the user agrees; no personal data in reports

7. Compliance, privacy and Google Play

Every item below is a release blocker: the app does not go to closed testing until each one is met and listed as checked in `docs/PROGRESS.md`.

Requirement	How the app meets it
Privacy policy, terms of service, refund policy	In-app screens plus public web pages; Play needs the privacy policy URL
Cookie policy and consent banner	The app sets no cookies. A "data and storage notice" explains what is stored on the device; the first-launch screen asks about optional error reporting with equal "Allow" and "No thanks" buttons
Consent boxes	Unticked by default; the API stores the timestamp and policy version of each consent
Age consent	Sign up requires 18 or over, or a parent or guardian's permission
Minimal data collection	Name, email, country, language and an optional phone number only. Location is used in the foreground, on user action, and never stored on the server
Audited third-party SDKs	`docs/THIRD_PARTY.md` lists each package, licence, purpose and data sent; it feeds the Play Data safety form
No dark patterns	Equal-weight choices, no pre-ticked boxes, sign out and delete account reachable in two taps from Profile
No hidden fees	The app is free and takes no payments in this phase
No fake reviews or unsupported claims	Section 5 rules: no ratings without real reviews, counts and sizes from data, live weather or none

Requirement	How the app meets it
Accessibility	Alt text on every photo, labels on every icon button for TalkBack, text contrast of at least 4.5:1 (section 4), 48 dp touch targets, full keyboard and D-pad focus order, text scaling to 200%
Business details	Business details screen and Play store listing
Unsubscribe link in emails	MVP sends transactional email only; any later marketing email needs opt-in and an unsubscribe link
Licensed fonts and images	Fonts under SIL Open Font License; every image carries credit and licence in the admin
Data deletion	Delete account in the app and on a public web page, as Play requires for apps with account creation
Safety positioning	Safety and route screens state that ranger guidance and park rules come first (budget section 6)

Google Play checklist:

- [ ] Target API level 36, as required for new apps from 31 August 2026 (source)
- [ ] Signed Android App Bundle with Play App Signing; signing keys kept outside the repository
- [ ] Permissions limited to internet, foreground location and, only if voice search stays, the microphone, each requested at the moment of use
- [ ] Data safety form completed from `docs/THIRD_PARTY.md`
- [ ] Account deletion link added in Play Console
- [ ] Store listing: screenshots, short and full description without unsupported claims, privacy policy URL
- [ ] Closed test: a personal developer account created after 13 November 2023 needs at least 12 testers opted in for 14 continuous days before production; organisation accounts are exempt (source)

Confirm during the budgeted legal review (KES 40,000) whether ISTA must register with Kenya's Office of the Data Protection Commissioner under the Data Protection Act, 2019.

8. Build phases

Nine phases, each ending in a sign-off; phases 2, 3 to 7 and 8 line up with the budget's release stages B (prototype), C (MVP) and D (field pilot).

1. **Phase 0: Setup.** File tree, package list with licences, `flutter create` with the approved application ID, Django project skeleton, CI running analyze and tests.
   - Done when: an empty app runs on an emulator, the backend starts, `flutter analyze` is clean and CI is green.
2. **Phase 1: Design system.** Theme, colour tokens (with the contrast fixes), bundled fonts, buttons, inputs, chips, cards, list rows, bottom bar, custom icons, debug design gallery.
   - Done when: gallery screenshots sit beside the web reference at 390 px and the owner approves them.
3. **Phase 2: Clickable prototype (budget stage B).** Every screen in section 5 with seed data from a local repository; no backend.
   - Done when: every screen is reachable, matches the design, and every icon button has a TalkBack label.
4. **Phase 3: Backend.** Models, admin with the publishing rules, API endpoints, OpenAPI schema, seed command for 10 to 20 curated records, tests.
   - Done when: an admin can create, verify and publish a place, and API tests pass.
5. **Phase 4: Accounts and data.** Sign up, sign in, password reset, email verification, profile, saved places, SQLite cache, consent records, delete account.
   - Done when: a test user can sign up, save a place, go offline, still see it, and delete the account with all personal data removed.
6. **Phase 5: Maps, location and directions.** Map tiles, pins, place selection, GPS, route request through the API, curated routes, offline straight-line fallback.
   - Done when: on a real Android phone, a route from Voi to Taita Hills draws correctly and offline mode shows a clear fallback.
7. **Phase 6: Offline packs and audio guides.** Pack download, resume, checksum, update, delete, storage display; audio player with transcript.
   - Done when: with airplane mode on, the Tsavo pack shows places, photos, safety info and plays audio.
8. **Phase 7: Compliance, languages and accessibility.** Legal screens, first-launch notice, error reporting consent, English and Kiswahili strings, contrast and TalkBack pass, text scaling at 200%.

    - Done when: every row in section 7 is checked and recorded in `docs/PROGRESS.md`.
9. **Phase 8: Release preparation (budget stage D).** Signed release bundle, store listing, Data safety form, closed test, field GPS and offline trials, crash monitoring.
    - Done when: the closed test has run 14 days with at least 12 testers (if the account needs it) and the field trial issues are fixed or logged.

9. Open items for the owner

Claude Code needs these answers at the phase shown; until then it uses a clearly labelled placeholder and records it in `docs/PROGRESS.md`.

Item	Needed by	Notes
App name on Play and application ID (for example `com.istasafaris.app`)	Phase 0	Cannot change after the first upload
Developer account type: personal or organisation	Phase 0	Decides whether the 12 tester closed test applies
Languages for launch	Phase 1	Note assumes English and Kiswahili
Map tile provider and offline tile rights	Phase 5	MapTiler or Google; written confirmation before any offline map promise
Routing provider for in-app directions	Phase 5	Options include a self-hosted OSRM server or a paid routing API
Weather: paid Open-Meteo plan or remove the weather line	Phase 5	Free tier is non-commercial only
Hosting for Django, PostgreSQL and media	Phase 3	Budget allows KES 42,000, 30,000 and 30,000 a year
Email and error reporting providers	Phase 4	Budget allows KES 18,000 a year for both plus monitoring
Verified attraction list with coordinates (10 to 20 records)	Phase 3	Confirm whether Mudanda Rock and Lion Ridge are real, verified places

Item	Needed by	Notes
Original photos, logo and audio narration with usage rights	Phase 6	Current images are screen captures
Who approves safety content and park routes	Phase 3	Name the role that signs off in the admin
Legal text: business name, registration, address, contacts, lawyer review	Phase 7	Web reference pages use `[PLACEHOLDER]` text
Public domain for the privacy policy and deletion pages	Phase 8	Play requires both URLs

Sources: ISTA Safaris full project budget (uploaded file, 25 October 2026) · Google Play target API requirements · Google Play testing requirements for new personal accounts · Open-Meteo terms of use